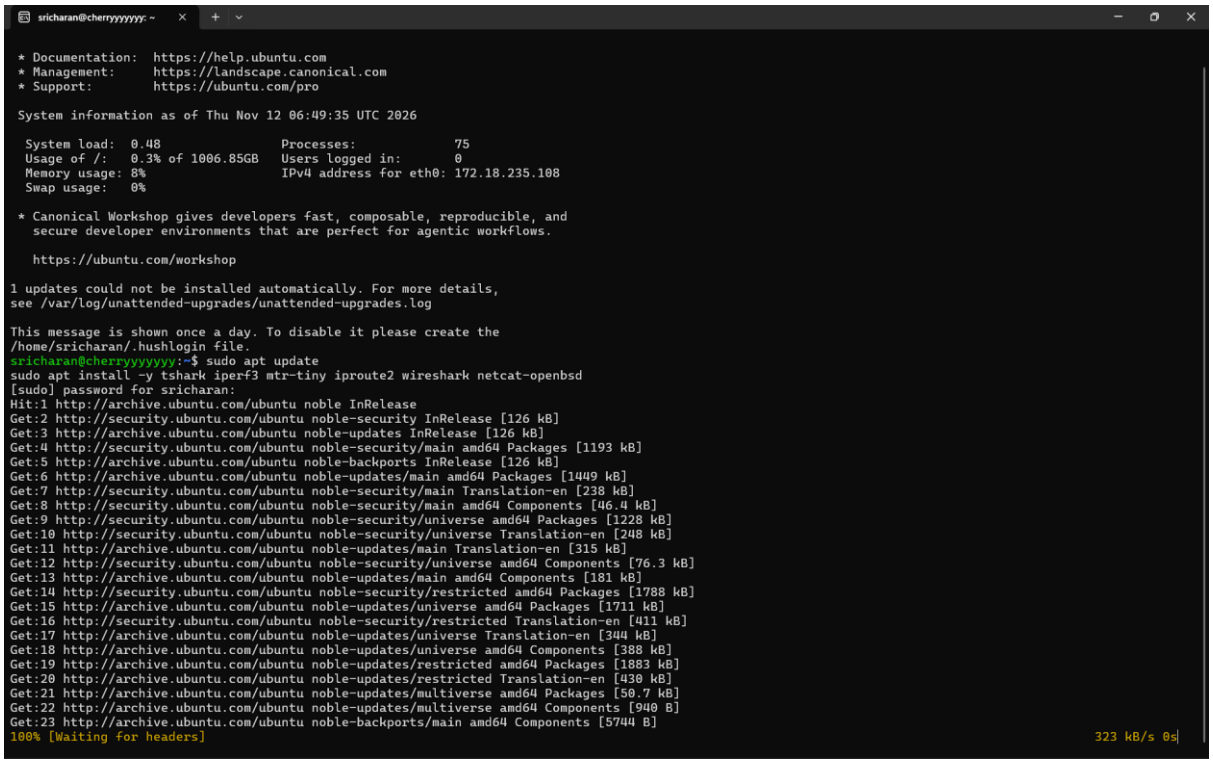
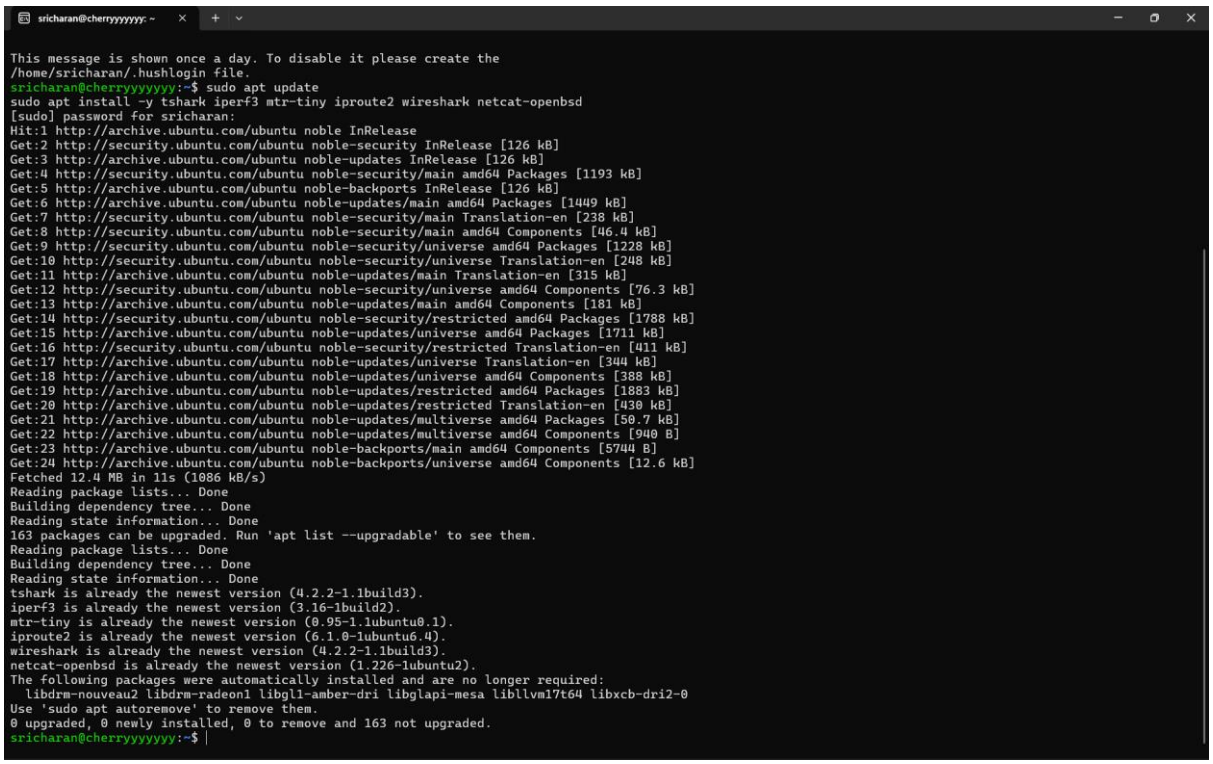
SKILL-2:

2520090019



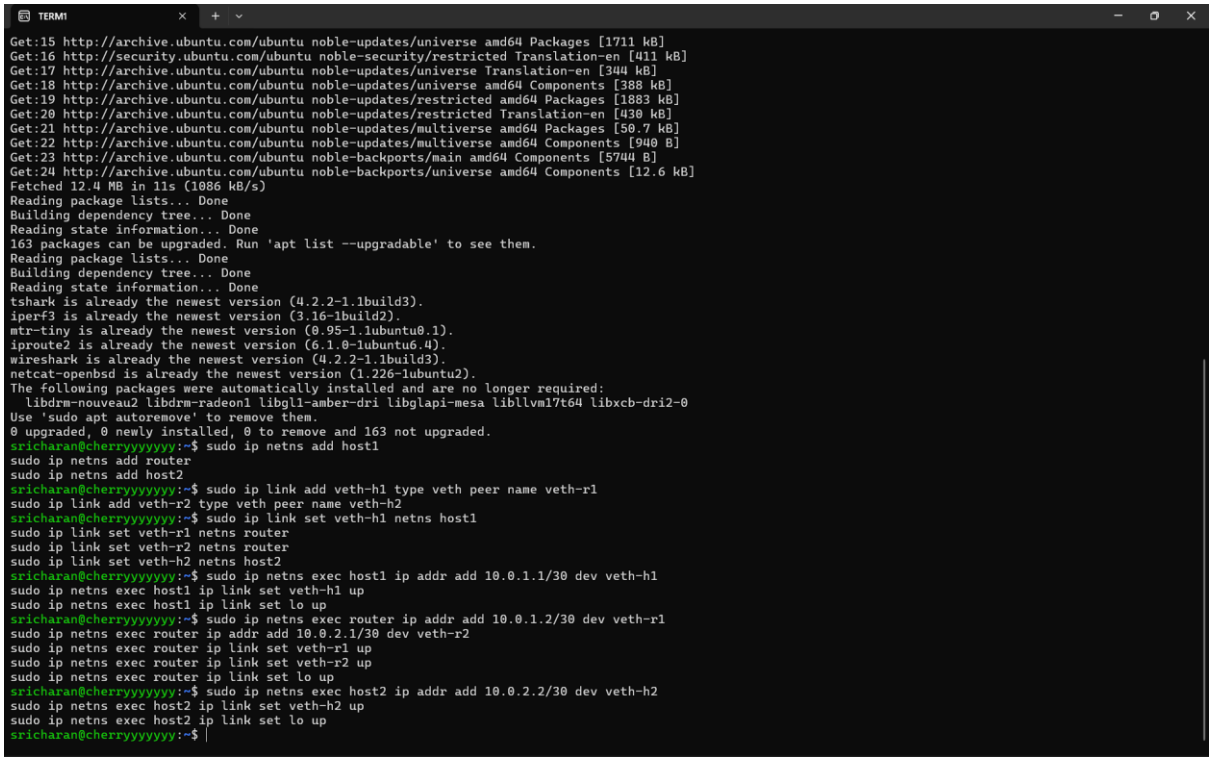


```
sricharan@cherryyyyyyy: ~
sricharan@cherryyyyyyy:~$ sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
[sudo] password for sricharan:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1193 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1449 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [238 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1228 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [248 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [315 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1788 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1711 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [411 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [344 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1883 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [430 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [50.7 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5744 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 12.4 MB in 11s (1086 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
163 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tshark is already the newest version (4.2.2-1.1build3).
iperf3 is already the newest version (3.16-1build2).
mtr-tiny is already the newest version (0.95-1.1ubuntu0.1).
iproute2 is already the newest version (6.1.0-1ubuntu6.4).
wireshark is already the newest version (4.2.2-1.1build3).
netcat-openbsd is already the newest version (1.226-1ubuntu2).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 163 not upgraded.
sricharan@cherryyyyyyy:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
sricharan@cherryyyyyyy:~$
```

```
sricharan@cherryyyyyyy:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
sricharan@cherryyyyyyy:~$
```

```
sricharan@cherryyyyyyy:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
sricharan@cherryyyyyyy:~$
```

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
sricharan@cherryyyyyyy:~$
```

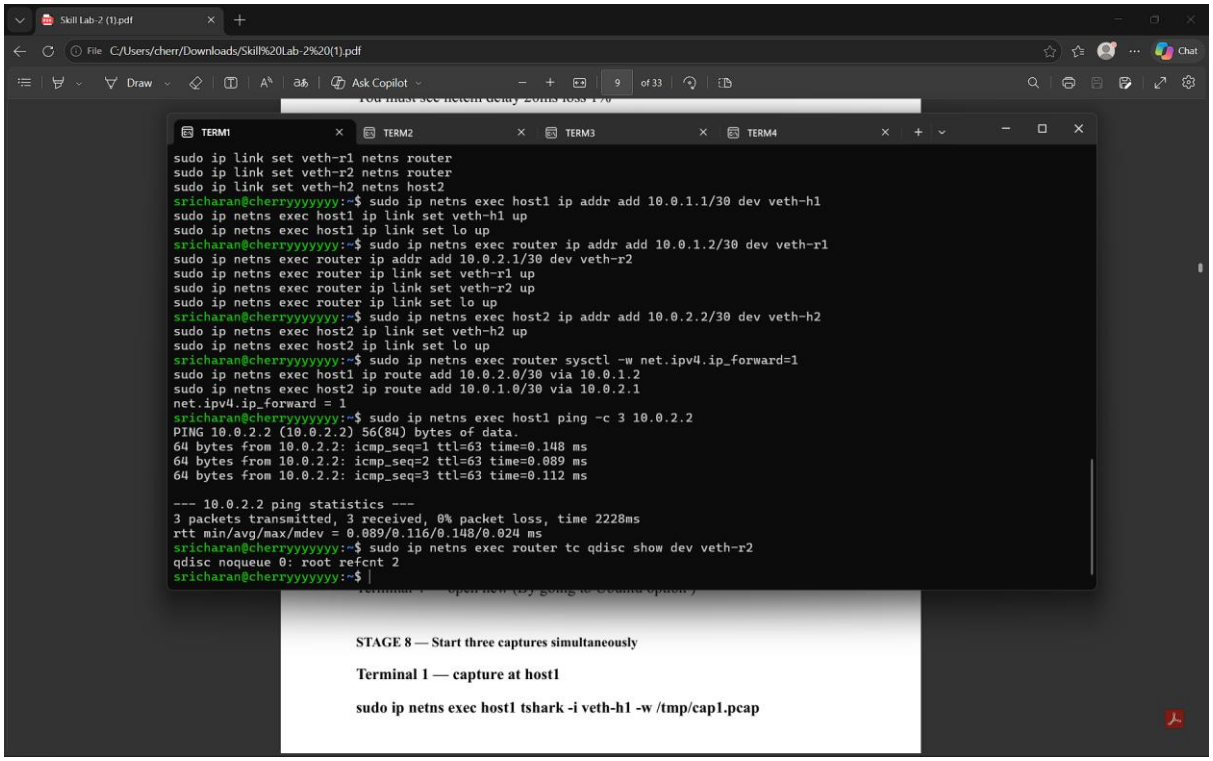



```
sricharan@cherryyyyyyy:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
sricharan@cherryyyyyyy:~$
```

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.148 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.089 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.112 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2228ms
rtt min/avg/max/mdev = 0.089/0.116/0.148/0.024 ms
sricharan@cherryyyyyyy:~$
```

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
sricharan@cherryyyyyyy:~$
```

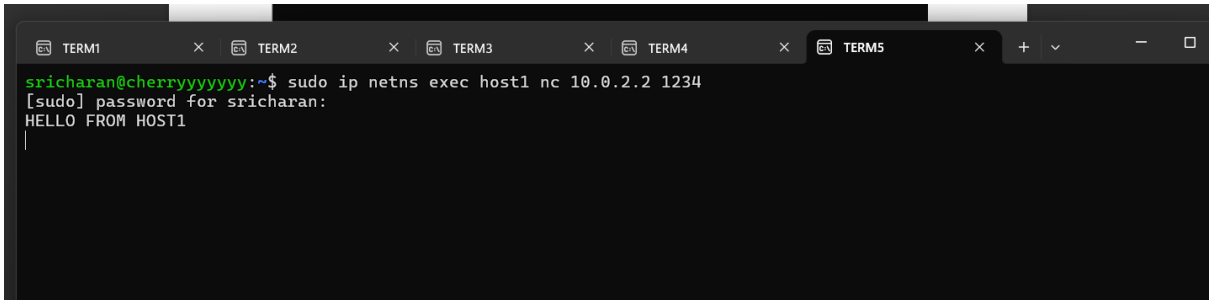


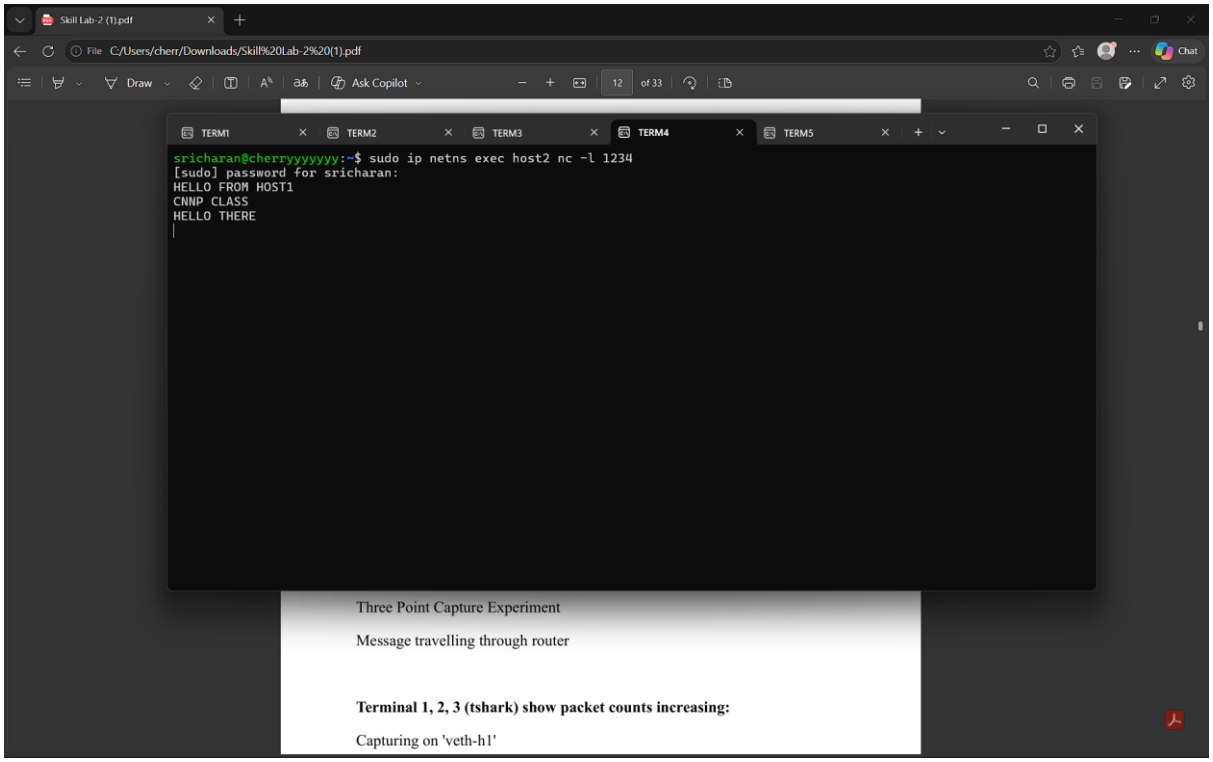

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
2
```

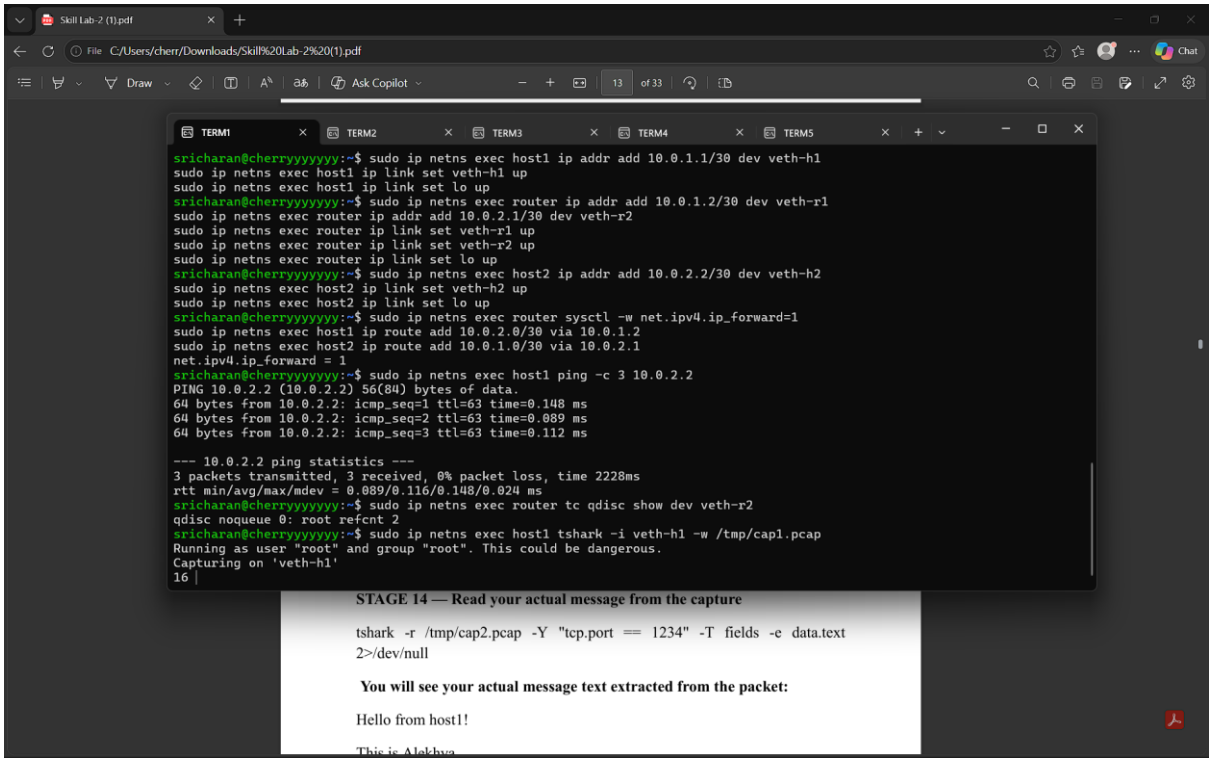
```
TERM1  TERM2  TERM3  TERM4
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
2
```

```
TERM1  TERM2  TERM3  TERM4
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
2
```

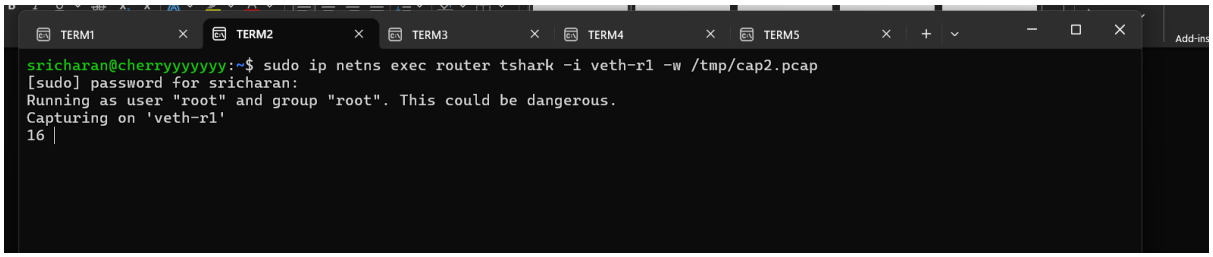
```
TERM1  TERM2  TERM3  TERM4
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 nc -l 1234
[sudo] password for sricharan:
```

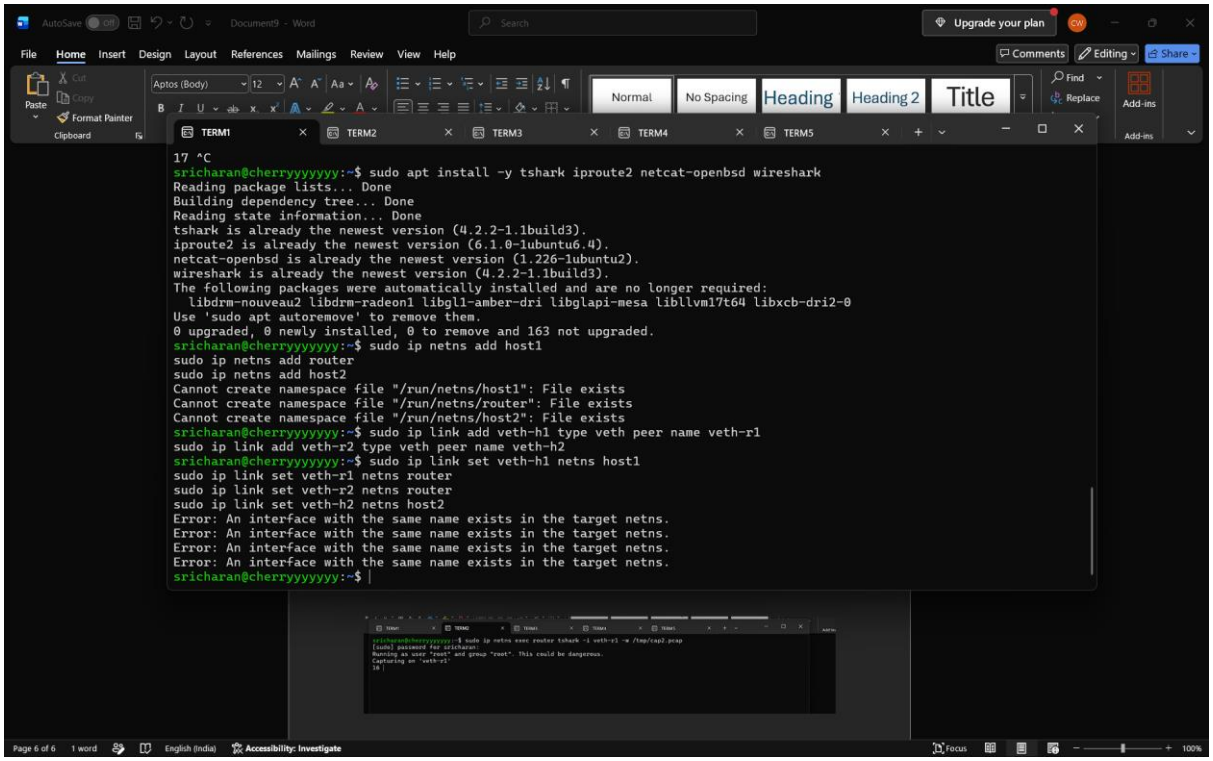

TERM1
TERM2
TERM3
TERM4
TERM5
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo] password for sricharan:
HELLO FROM HOST1

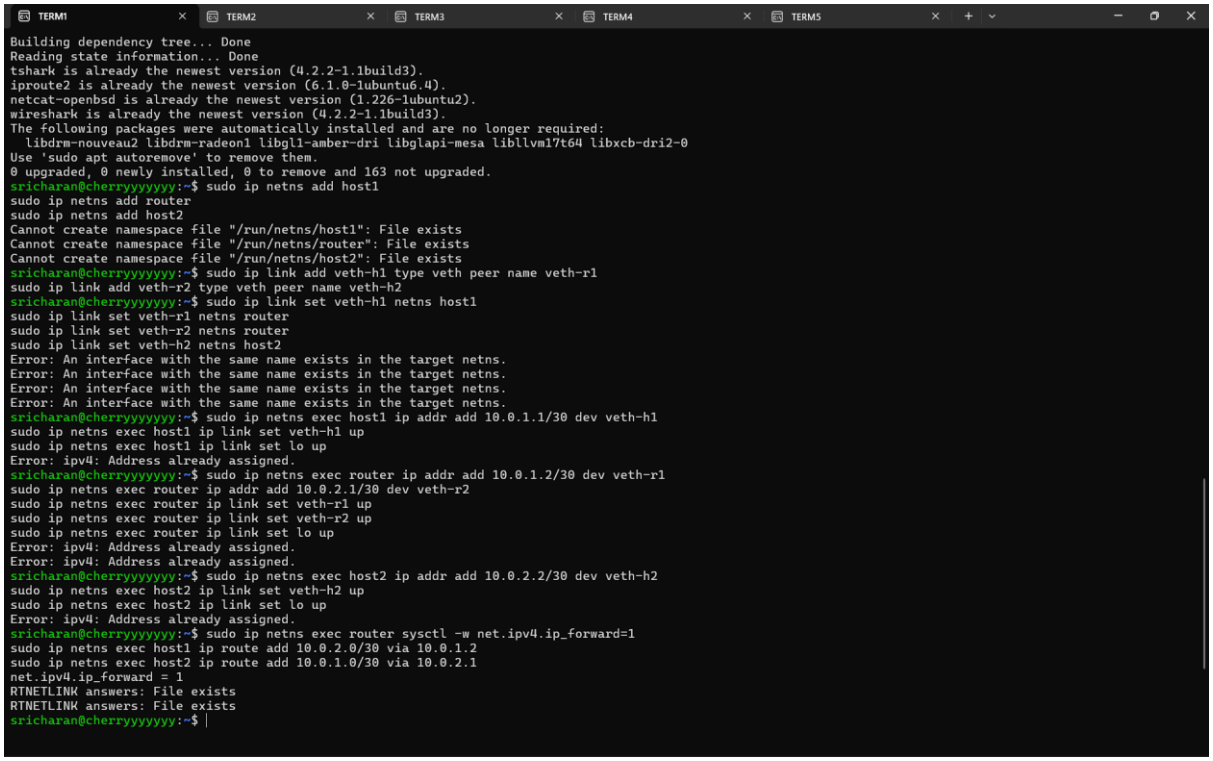

Skill Lab-2 (1).pdf
File C:/Users/cherr/Downloads/Skill%20Lab-2%20(1).pdf
Draw
Ask Copilot
12 of 33
Chat
TERM1
TERM2
TERM3
TERM4
TERM5
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 nc -l 1234
[sudo] password for sricharan:
HELLO FROM HOST1
CNNP CLASS
HELLO THERE
Three Point Capture Experiment
Message travelling through router
Terminal 1, 2, 3 (tshark) show packet counts increasing:
Capturing on 'veth-h1'

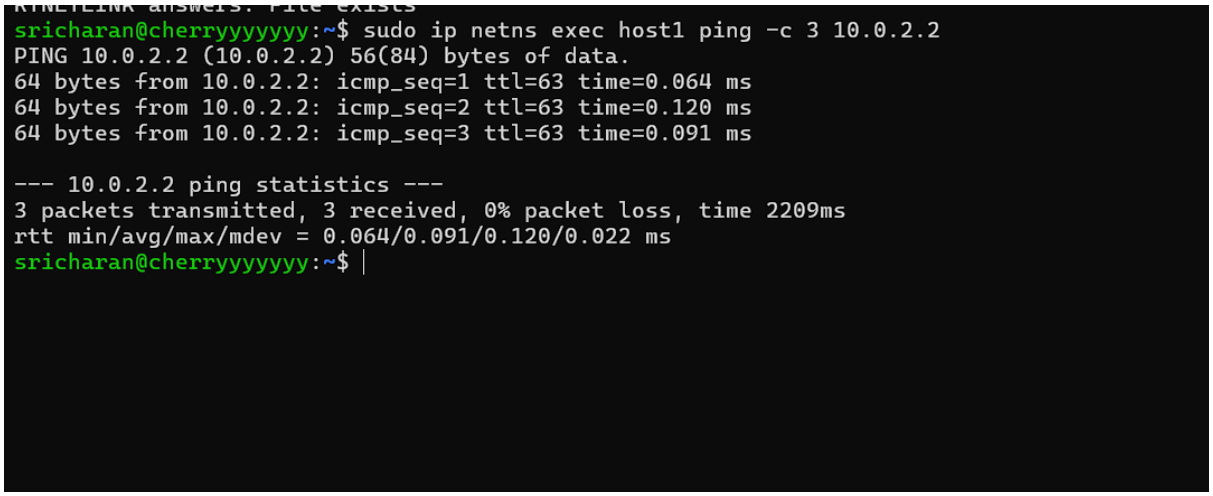

Skill Lab-2 (1).pdf
File C:/Users/cherr/Downloads/Skill%20Lab-2%20(1).pdf
Draw
Ask Copilot
13 of 33
Chat
TERM1
TERM2
TERM3
TERM4
TERM5
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
sricharan@cherryyyyyyy:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
sricharan@cherryyyyyyy:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.148 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.089 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.112 ms
--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2228ms
rtt min/avg/max/mdev = 0.089/0.116/0.148/0.024 ms
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
16
STAGE 14 — Read your actual message from the capture
tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null
You will see your actual message text extracted from the packet:
Hello from host1!
This is Alekhya

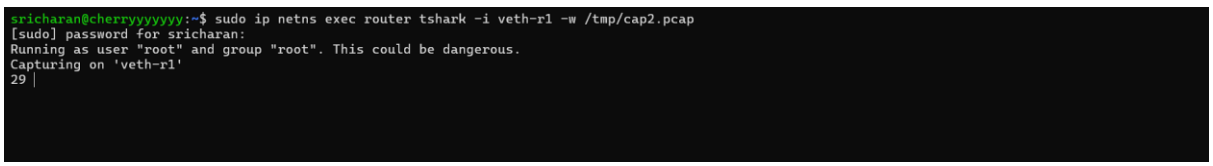
```
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
17
```

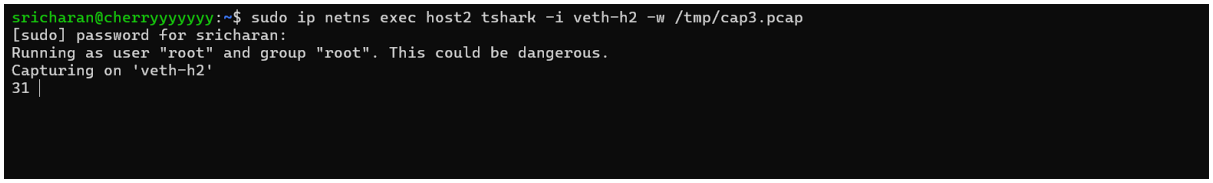





TERM1
TERM2
TERM3
TERM4
TERM5
Building dependency tree... Done
Reading state information... Done
tshark is already the newest version (4.2.2-1.1build3).
iproute2 is already the newest version (6.1.0-1ubuntu6.4).
netcat-openbsd is already the newest version (1.226-1ubuntu2).
wireshark is already the newest version (4.2.2-1.1build3).
The following packages were automatically installed and are no longer required:
libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 163 not upgraded.
sricharan@cherryyyyyyy:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
sricharan@cherryyyyyyy:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
sricharan@cherryyyyyyy:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Error: ipv4: Address already assigned.
sricharan@cherryyyyyyy:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
Error: ipv4: Address already assigned.
Error: ipv4: Address already assigned.
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
Error: ipv4: Address already assigned.
sricharan@cherryyyyyyy:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
sricharan@cherryyyyyyy:~$


RTNETLINK answers: File exists
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.064 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.120 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.091 ms
--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2209ms
rtt min/avg/max/mdev = 0.064/0.091/0.120/0.022 ms
sricharan@cherryyyyyyy:~$


sricharan@cherryyyyyyy:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
29


sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
31

TERM1 TERM2 TERM3 TERM4 TERM5

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 nc -l 1234
[sudo] password for sricharan:
HELLO FROM HOST1
CNNP CLASS
HELLO THERE
```

TERM1 TERM2 TERM3 TERM4 TERM5

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
29 ^C
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

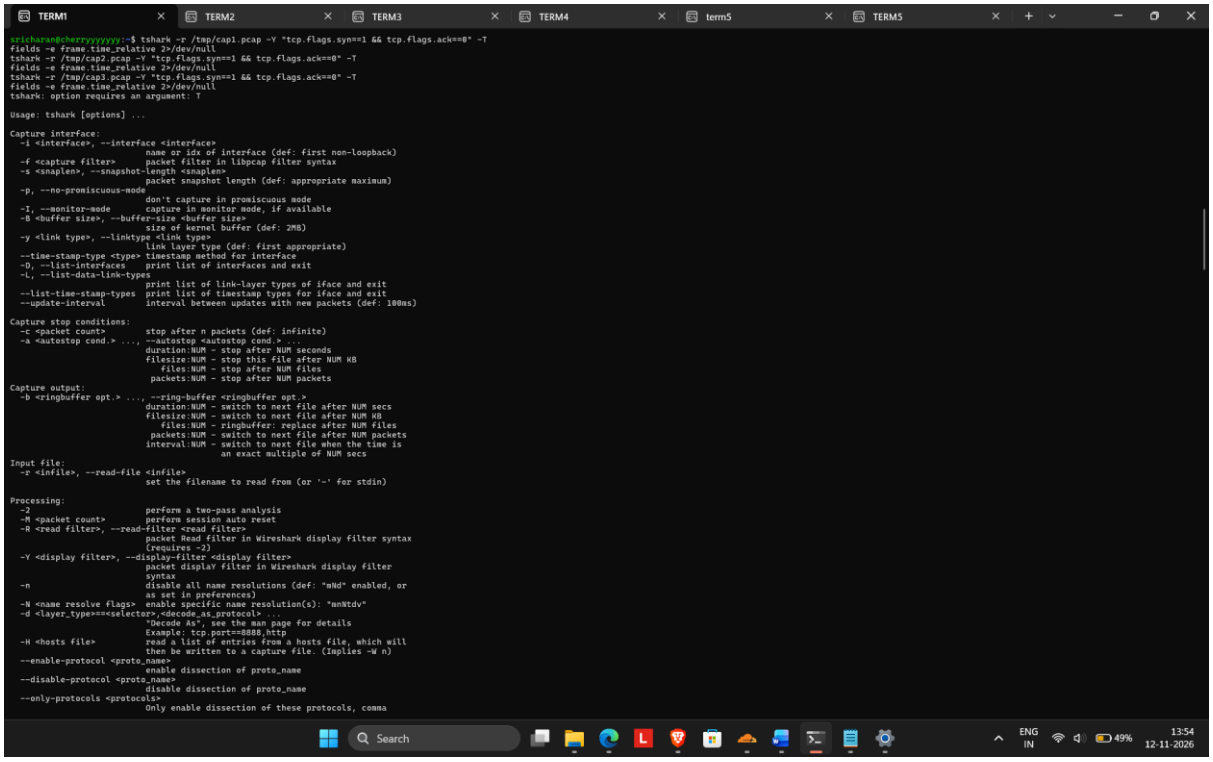
TERM1 TERM2 TERM3 TERM4 TERM5

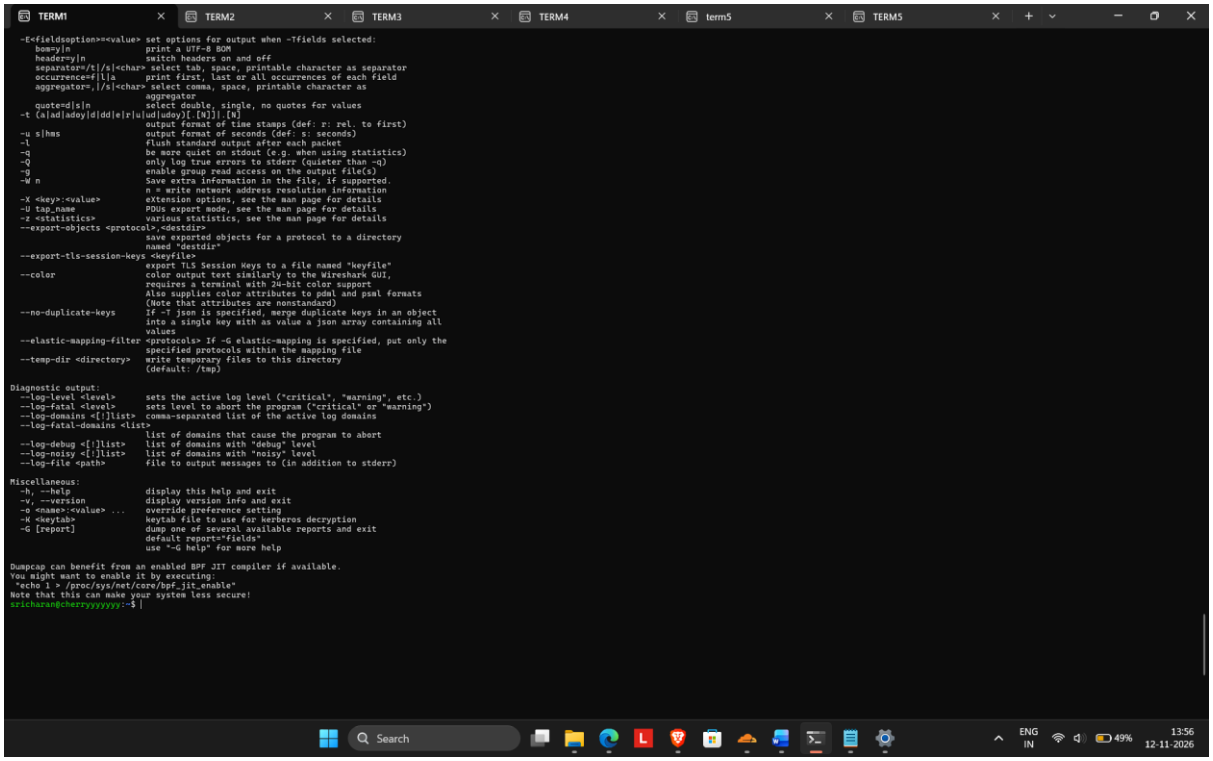
```
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
31 ^C
sricharan@cherryyyyyyy:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

```
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.064 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.120 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.091 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2209ms
rtt min/avg/max/mdev = 0.064/0.091/0.120/0.022 ms
sricharan@cherryyyyyyy:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss
1%
Command line is not complete. Try option "help"
1%: command not found
sricharan@cherryyyyyyy:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
[sudo] password for sricharan:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
13 ^C
sricharan@cherryyyyyyy:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw------- 1 root root 1.6K Nov 12 08:17 /tmp/cap1.pcap
-rw------- 1 root root 1.6K Nov 12 08:17 /tmp/cap2.pcap
-rw------- 1 root root 1.6K Nov 12 08:17 /tmp/cap3.pcap
sricharan@cherryyyyyyy:~$
```

```
sricharan@cherryyyyyyy:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
sricharan@cherryyyyyyy:~$
```




TERM1
TERM2
TERM3
TERM4
term5
TERM5
-E<fieldsoption>=<value> set options for output when -Tfields selected:
bom=y|n print a UTF-8 BOM
header=y|n switch headers on and off
separator=/t|/s|<char> select tab, space, printable character as separator
occurrence=f|l|a print first, last or all occurrences of each field
aggregator=,|/s|<char> select comma, space, printable character as
aggregator
quote=d|s|n select double, single, no quotes for values
-t (a|ad|adoy|d|dd|e|r|u|ud|udoy)[.[N]][.(N)
output format of time stamps (def: r: rel. to first)
-u s|hms output format of seconds (def: s: seconds)
-l flush standard output after each packet
-q be more quiet on stdout (e.g. when using statistics)
-Q only log true errors to stderr (quieter than -q)
-g enable group read access on the output file(s)
-W n Save extra information in the file, if supported.
n = write network address resolution information
-X <key>:<value> eXtension options, see the man page for details
-U tap_name PDUs export mode, see the man page for details
-z <statistics> various statistics, see the man page for details
--export-objects <protocol>,<destdir>
save exported objects for a protocol to a directory
named "destdir"
--export-tls-session-keys <keyfile>
export TLS Session Keys to a file named "keyfile"
--color color output text similarly to the Wireshark GUI,
requires a terminal with 24-bit color support
Also supplies color attributes to pdml and psml formats
(Note that attributes are nonstandard)
--no-duplicate-keys If -T json is specified, merge duplicate keys in an object
into a single key with as value a json array containing all
values
--elastic-mapping-filter <protocols> If -G elastic-mapping is specified, put only the
specified protocols within the mapping file
--temp-dir <directory> write temporary files to this directory
(default: /tmp)
Diagnostic output:
--log-level <level> sets the active log level ("critical", "warning", etc.)
--log-fatal <level> sets level to abort the program ("critical" or "warning")
--log-domains <[!]list> comma-separated list of the active log domains
--log-fatal-domains <list>
list of domains that cause the program to abort
--log-debug <[!]list> list of domains with "debug" level
--log-noisy <[!]list> list of domains with "noisy" level
--log-file <path> file to output messages to (in addition to stderr)
Miscellaneous:
-h, --help display this help and exit
-v, --version display version info and exit
-o <name>:<value> ... override preference setting
-K <keytab> keytab file to use for kerberos decryption
-G [report] dump one of several available reports and exit
default report="fields"
use "-G help" for more help
Dumpcap can benefit from an enabled BPF JIT compiler if available.
You might want to enable it by executing:
"echo 1 > /proc/sys/net/core/bpf_jit_enable"
Note that this can make your system less secure!
sricharan@cherryyyyyyy:~$
Search
ENG
IN
49%
13:56
12-11-2026